Product Name: DUAL GOLD HERBICIDE
APVMA Approval No: 50477/133889

Label Name:	DUAL GOLD HERBICIDE

Signal Headings:	CAUTION KEEP OUT OF REACH OF CHILDREN READ SAFETY DIRECTIONS BEFORE OPENING OR USING

Constituent Statements:	960 g/L S-METOLACHLOR

Mode of Action:	GROUP 15 HERBICIDE

Statement of Claims:	Controls certain annual grasses and broadleaf weeds in certain crops as specified in the Directions for Use

Net Contents:	5 L - 1000 L

Restraints:	DO NOT apply by aircraft DO NOT apply to waterlogged soils DO NOT apply if heavy rains or storms that are likely to cause runoff are forecast within 2 days of application DO NOT irrigate to the point of runoff for at least 2 days after application Fallows, Mungbeans, Sorghum: DO NOT apply more than 2 L/ha in a single season Mungbeans: DO NOT apply after the 4 leaf growth stage Sorghum: DO NOT apply after the 6 leaf growth stage. Spray drift restraints Specific definitions for terms used in this section of the label can be found at apvma.gov.au/spraydrift. DO NOT allow bystanders to come into contact with the spray cloud. DO NOT apply in a manner that may cause an unacceptable impact to native vegetation,

	agricultural crops, landscaped gardens and aquaculture production, or cause contamination of plant or livestock commodities, outside the application site from spray drift. Wherever possible, correctly use application equipment designed to reduce spray drift and apply when the wind direction is away from these sensitive areas. DO NOT apply unless the wind speed is between 3 and 20 kilometres per hour at the application site during the time of application. DO NOT apply if there are hazardous surface temperature inversion conditions present at the application site during the time of application. Surface temperature inversion conditions exist most evenings 1 to 2 hours before sunset and persist until 1 to 2 hours after sunrise.

Directions for Use:	This section contains file attachment.

Other Limitations:	

Withholding Periods:	Canola, Mustard: DO NOT GRAZE TREATED FIELDS FOR 10 WEEKS AFTER APPLICATION Green Beans, Navy Beans: DO NOT HARVEST BEANS FOR 8 WEEKS AFTER APPLICATION DO NOT GRAZE OR CUT FOR STOCK FOOD FOR 10 WEEKS AFTER APPLICATION Herbs, Chervil, Mizuna, Rucola, Lemon Verbena: DO NOT HARVEST FOR 8 WEEKS AFTER APPLICATION Galangal, Turmeric: DO NOT HARVEST FOR 23 WEEKS AFTER APPLICATION Mungbeans: DO NOT GRAZE OR CUT FOR STOCK FOOD FOR 6 WEEKS AFTER APPLICATION Pastures: DO NOT GRAZE OR CUT FOR STOCK FOOD FOR 13 WEEKS AFTER APPLICATION Sorghum: DO NOT GRAZE OR CUT FOR STOCK FOOD FOR 4 WEEKS AFTER APPLICATION Sweet Potatoes: DO NOT HARVEST, GRAZE OR CUT FOR STOCK FOOD FOR 23 WEEKS AFTER APPLICATION Wheat, Barley, Oats, Triticale, Spinach, Silverbeet: DO NOT GRAZE OR CUT FOR STOCK FOOD FOR 8 WEEKS AFTER APPLICATION HARVEST withhold for all other crops: NOT REQUIRED WHEN USED AS DIRECTED FORAGE withhold for all other crops including Fallow (except Sweet Potatoes): DO NOT GRAZE OR CUT FOR STOCK FOOD FOR 13 WEEKS AFTER APPLICATION

Trade Advice:	EXPORT OF TREATED PRODUCE: Growers should note that maximum residue limits (MRLs) or import tolerances may not exist in all markets for edible produce treated with this product. If you are growing edible produce for export, please check with industry representative or Syngenta Australia Pty Ltd for the latest information on MRLs and import tolerances before using this product.

General Instructions:	DUAL GOLD Herbicide is a short residual, pre-emergent herbicide registered for the control of many important annual grasses and broadleaf weeds in the crops nominated. The product does not control the important broadleaf weeds in maize, sweet corn and CONCEP II or EPIVIO C treated sorghum but, when mixed with atrazine, control of many important annual weeds and grasses is obtained without the problem of carryover of herbicides affecting succeeding crops. The product acts mainly through shoot (coleoptile) absorption. Its effectiveness depends on the occurrence of rainfall or irrigation to move it down into the

	top 3 to 4 cm of soil, where it must be present at the time of weed germination. Duration and effectiveness of weed control will depend on use rate, soil type and rainfall or irrigation after application. Good soil moisture to ensure soil movement and uptake by emerging weed seeds. Drying of the top soil can result in less effective weed control. Mixing Fill the spray tank to one quarter full. Then add DUAL GOLD while adding water to make up the final spray volume. Pour DUAL GOLD into the stream of incoming water. Good agitation at the time DUAL GOLD is added to the tank will ensure good mixing. When Tank Mixing: Wettable powder or water dispersible granule formulations should be added to the tank first, followed by suspension concentrates (flowables), water soluble salts then DUAL GOLD or other emulsifiable concentrate formulations. Maintain thorough agitation during mixing and application. Agitate tank mixes vigorously if allowed to stand. Note: Tank mix spray solution should NOT be left standing in the vat overnight. Read and follow all label directions, restraints, plant-back periods, withholding periods, resistance management, precautions, protection statements and safety directions for the tank mix products. Application Ground Application: Apply in a minimum of 60 L/ha of water Compatibility DUAL GOLD is compatible with Cotogard* SC, Cotoran* SC, GESAGARD 500 SC, GESAPRIM Granules, LOGRAN, Basagran*, SPRAY.SEED, Roundup and Weedmaster DST. Crop Replanting Interval DO NOT plant crops other than broccoli, brussel sprouts, cabbages, cauliflowers, cotton, maize, peanuts, soybeans, sunflowers, sweet corn, tobacco or CONCEP II or EPIVIO C treated sorghum within 6 months of application of DUAL GOLD. To Avoid Crop Damage As metolachlor is known to cause some crop damage in sandy-sandy loam soils with low organic matter, it is recommended that growers test the safety of metolachlor on a small area of their crop before using the product on larger areas. If any adverse crop symptoms are observed within 14 days of application in which irrigation or rainfall has occurred, then the product should not be used. WARNING - Cotton: DO NOT use when planting into dry soil or soil with marginal moisture. Under dry planting conditions, heavy rain or flood irrigation soon after planting, together with below average soil temperatures, can result in crop injury due to the inability of the germinating seedling to metabolise above normal amounts of the herbicide that may accumulate in the root zone as a result of accelerated capillary movement. DUAL GOLD should not be used in situations where these conditions are likely to occur. WARNING - Brassica Crops: Crop retardation may occur where DUAL GOLD is used on soils that have a low organic matter content and which contain more than 60% of fine sand and silt. On these soil types, use rates of DUAL GOLD towards the lower end of the rate range and apply onto moist soil. Irrigation after application should be limited to no more than 25 mm. DO NOT use more than once per year. Zero Tillage (Canola, Barley, Oats, Wheat, Triticale): When sowing using knife points and zero tillage, and leaving a seeding slot, DO NOT apply post-seeding unless harrows (or similar) have filled-in the seeding slot before application. Alternatively, apply DUAL GOLD immediately before seeding.

Resistance Warning:	Resistant Weeds Warning - GROUP 15 HERBICIDE DUAL GOLD Herbicide is a member of the α-chloroacetamides group of herbicides. The product has the inhibition of very long chain fatty acid synthesis mode of action.

	For weed resistance management, the product is a Group 15 herbicide. Some naturally occurring weed biotypes resistant to the product and other Group 15 herbicides may exist through normal genetic variability in any weed population. The resistant individuals can eventually dominate the weed population if these herbicides are used repeatedly. These resistant weeds will not be controlled by this product or other Group 15 herbicides. Since the occurrence of resistant weeds is difficult to detect prior to use, Syngenta Australia Pty Ltd accepts no liability for any losses that may result from the failure of this product to control the resistant weeds. Advice as to strategies and alternative treatments that can be used should be obtained from your local supplier, consultant, local Department of Agriculture, Primary Industries Department or a Syngenta representative. Large numbers of healthy surviving weeds can be an indication that resistance is developing. Efforts should be made to prevent seed set of the surviving weeds. DO NOT exceed the maximum recommended rate of DUAL GOLD in any year. DO NOT apply other Group 15 herbicides with the inhibition of very long chain fatty acid synthesis mode of action to a crop in the same season. If the user suspects that the target weed population is resistant to herbicides with this mode of action, DUAL GOLD or other Group 15 herbicides should not be used. Strategies to minimise the risk of herbicide resistance are available. The above recommendations should be incorporated into an Integrated Weed Management (IWM) Program that uses a diverse range of options to manage grass weeds and avoids over reliance on any one method of control. Consult your farm chemical supplier, consultant, local Department of Agriculture or Primary Industries, or local Syngenta representative for details. Integrated Weed Management for over-the-top use on Roundup Ready Flex cotton It is advised that consultation on Integrated Weed Management be undertaken with an accredited agronomist or program prior to use of DUAL GOLD (alone or in a tank mix) over Roundup Ready Flex cotton. Please refer to the label of the tank mix partner (eg Weedmaster DST) for additional information on weed resistance management.

Precautions:	Re-entry Period DO NOT allow entry into treated areas until the spray has dried.

Protections:	PROTECTION OF LIVESTOCK DO NOT graze or cut Sweet Potato leaves for stock food. PROTECTION OF CROPS, NATIVE AND OTHER NON-TARGET PLANTS DO NOT drain or flush equipment on or near desirable trees or other plants, where their roots may extend, or in situations where by movement of soil or by seepage absorption of the herbicide may occur. PROTECTION OF WILDLIFE, FISH, CRUSTACEANS AND ENVIRONMENT TOXIC TO FISH. DO NOT contaminate streams, rivers or waterways with the chemical or used containers. DO NOT apply under meteorological conditions or from spraying equipment which could be expected to cause spray to drift onto adjacent areas, particularly wetlands, waterbodies or watercourses.

Storage and Disposal:	STORAGE AND DISPOSAL Store in the closed, original container in a dry, cool, well ventilated area out of direct sunlight. Non-refillable containers: Triple rinse containers before disposal. Add rinsings to spray tank. DO NOT dispose of undiluted chemicals on site. If recycling, replace cap and return clean containers to recycler or designated collection point. If not recycling, break, crush or puncture and and deliver empty packaging to an approved waste management facility. If an approved waste management facility is not available, bury the empty packaging 500 mm

	below the surface in a disposal pit specifically marked and set up for this purpose, clear of waterways, desirable vegetation and tree roots, in compliance with relevant Local, State or Territory government regulations. DO NOT burn empty containers or product. Refillable containers: Empty contents fully into application equipment. Close all valves and return to point of supply for refill or storage.

Safety Directions:	Will damage the eyes. Repeated exposure may cause allergic disorders. Sensitive workers should use protective clothing. Avoid contact with eyes. When opening the container and preparing spray wear: • cotton overalls buttoned to the neck and wrist • a washable hat • elbow-length PVC gloves and • goggles. When using the prepared spray, wear: • cotton overalls buttoned to the neck and wrist • a washable hat • elbow-length PVC gloves and • face shield or goggles. If product in eyes, wash it out immediately with water. Wash hands after use. After each day's use, wash gloves, face shield or goggles and contaminated clothing.

First Aid Instructions:	If poisoning occurs, contact a doctor or Poisons Information Centre. Phone 131 126.

First Aid Warnings:	

DIRECTIONS FOR USE

Crops	Weeds Controlled	State	Rate	Critical Comments
Barley, Oats	**Suppression** of Annual Ryegrass	NSW, Vic, Tas, SA, WA, ACT only	375 to 500 mL/ha	Apply to moist soils immediately after planting and before crop and weeds emerge. Use the lower rate on light soils. Application should be made to a moist seed bed and sufficient rain to thoroughly wet the top 3 to 4 cm of soil should occur within 10 day after spraying. Damage may occur if crop is sown too shallow (less than 4 cm). Damage can occur on light sandy soil types after heavy rain. See GENERAL INSTRUCTIONS for use in Zero-Tillage Systems.
Broccoli, Brussels Sprouts, Cabbages, Cauliflowers	Barnyard Grass, Blackberry Nightshade, Chickweed (**suppression**), Deadnettle, Fat Hen, Feathertop Rhodes Grass, Love Grass, Pigeon Grass, Pigweed (**suppression**), Potato Weed, Shepherd's Purse, Stinging Nettle, Summer Grass, Wireweed (**suppression**)	Qld, NSW, Vic, Tas, NT only	1.5 to 2 L/ha	Apply immediately after transplanting. Sufficient irrigation to wet the top 3 to 4 cm of soil should be applied within 24 hours. Use rates towards the higher end of the range where Blackberry Nightshade, Chickweed, Fat Hen, Love Grass, Pigeon Grass, Pigweed, Potato Weed and Wireweed are the major problem. See GENERAL INSTRUCTIONS for Warning - Brassica Crops.
	Crowsfoot Grass, Fierce Thornapple (**suppression**), Liverseed Grass, Winter Grass	NSW only		Apply immediately after transplanting. Sufficient irrigation to wet the top 3 to 4 cm of soil should be applied within 24 hours. Use rates towards the higher end of the range where Apple of Peru, Common Sowthistle, Fierce Thornapple, Liverseed Grass, Redroot Amaranth, Twiggy Turnip and Nettleleaf Goosefoot are the major problems. **WARNING: Qld only:** Stunting and reduced yield may occur in Broccoli in red soils and light alluvial soils. See GENERAL INSTRUCTIONS for Warning - Brassica Crops.
	Apple of Peru, Awnless Barnyard Grass, Fierce Thornapple (**suppression**), Liverseed Grass, Yellow Weed, Redroot Amaranth (**suppression**)	Qld only		
	Annual Ryegrass, Common Cotula, Common Sowthistle (**suppression**), Crowsfoot Grass, Fiddle Dock, Italian Ryegrass, Liverseed Grass, Nettleleaf Goosefoot (**suppression**), Twiggy Turnip (**suppression**), Winter Grass	Vic, Tas only		

Crops	Weeds Controlled	State	Rate	Critical Comments
Brassica Leafy Vegetables including Chinese White Cabbage, Chinese Flowering Cabbage, Chinese Broccoli, Indian Mustard Cabbage, Chinese Cabbage, Mizuna Greens, Red Mustard, Curled Mustard, Chinese Flat Cabbage, Mustard Greens, Turnip Greens, Rutabaga Greens and Mibuna Greens	Annual Ryegrass, Apple of Peru, Awnless Barnyard Grass, Blackberry Nightshade, Chickweed (**suppression**), Common Cotula, Common Sowthistle (**suppression**), Crowsfoot Grass, Deadnettle, Fat Hen, Feathertop Rhodes Grass, Fiddle Dock, Fierce Thornapple (**suppression**), Italian Ryegrass, Liverseed Grass, Love Grass, Nettleleaf Goosefoot (**suppression**), Pigeon Grass, Pigweed (**suppression**), Potato Weed, Redroot Amaranth (**suppression**), Shepherd's Purse, Stinging Nettle, Summer Grass, Twiggy Turnip (**suppression**), Winter Grass, Wireweed (**suppression**), Yellow Weed	All States	1 to 2 L/ha	Apply one treatment only, immediately after transplanting. Apply in a minimum of 60 L water per hectare. Apply sufficient irrigation after application to wet the upper 3 to 4 cm of soil within 24 hours. Use rates towards the higher end of the range where Blackberry Nightshade, Chickweed, Fat Hen, Love Grass, Pigweed, Potato Weed and Wire Weed are major problems (Refer to Brassica use pattern on product label). See GENERAL INSTRUCTIONS for Warning - Brassica Crops.
Culinary Herbs: Basil, Bay Leaves, Borage, Chives, Coriander, Dill, Fennel, Lemon Balm, Lemon Grass, Kaffir Lime Leaves, Marigold Flowers, Marjoram (Oregano), Mints, Nasturtium Leaves, Parsley, Rosemary, Sage, Salad Burnett, Sorrel, Tarragon, Thyme **Savoury Root Vegetable:** Galangal **Leafy Vegetables:** Rucola (Rocket), Chervil, Mizuna. **Teas:** Lemon Verbena **Spice:** Turmeric. **Edible Flowers:** Dianthus, Roses		All States	1.5 to 2 L/ha	Apply before, at, or immediately after planting/transplanting and before crops and weeds have germinated. Sufficient irrigation to wet the top 3 to 4 cm of soil should be applied within 24 hours. DO NOT use more than once per season.

Crops	Weeds Controlled	State	Rate	Critical Comments
Canola and Mustard (oilseed cultivars) (*Brassica juncea*)	Toad Rush	All States	0.15 to 0.25 L/ha	Apply at or immediately after planting and before crops and weeds emerge. Apply to moist soil. Use rates toward the higher end of the range where longer residual control is required. See GENERAL INSTRUCTIONS for use in Zero-Tillage Systems.
Cotton (Including Roundup Ready* Flex cotton and Liberty Link cotton)	Barnyard Grass, Crowsfoot Grass, Feathertop Rhodes Grass, Liverseed Grass, Love Grass, Pigeon Grass, Summer Grass, Trad (Wandering Jew)	Qld, NSW, WA only	1 L/ha	**Pre-emergent**: Apply before, at or immediately after planting and before crops and weeds have germinated using a low volume boom sprayer applying 60 to 120 L mixture/ha. Rain or irrigation is necessary within 10 days of spraying to thoroughly wet the top 3 to 4 cm of soil. If rain does not occur or irrigation is unavailable, incorporate chemical to a depth of 3 to 4 cm. DO NOT throw untreated soil onto treated areas as this will reduce weed control. DO NOT use on soils where the sand content exceeds 70%. Use lower rate where the sand content of the soil is within the range of 50% to 70%. See GENERAL INSTRUCTIONS for Warning - Cotton.
Cotton (Including Roundup Ready* Flex cotton and Liberty Link cotton)	Awnless Barnyard Grass (*Echinochloa colona*), Barnyard Grass (*Echinochloa crus-galli*), Dinebra (*Dinebra retroflexa*), Crowsfoot Grass, Liverseed Grass, Lovegrass, Pigeon Grass, Summer Grass, Trad (Wandering Jew)	Qld, NSW, WA only	1 L/ha	**Over the top or directed spray applications:** Apply as a standalone over-the-top or directed spray application to a weed-free soil surface, using a boom sprayer applying 85 to 120 L of mixture per hectare. Apply only one application of DUAL GOLD® per crop. Apply from the 4 node to 18 node crop growth stage. For maximum efficacy, ensure even distribution of spray across the furrow. Use a directed spray if shading by the crop is likely to affect herbicide placement. Some minor transient leaf spotting or leaf scorching may occur after application. To avoid any leaf spotting or leaf scorching use a directed or shielded spray. Rain or irrigation is necessary within 10 days of spraying to thoroughly wet the top 3 to 4 cm of soil. If rain does not occur or irrigation is unavailable to incorporate, reduced residual control may occur.
Roundup Ready Flex Cotton	Awnless Barnyard Grass (*Echinochloa colona*), Barnyard Grass (*Echinochloa crus-galli*), Dinebra (*Dinebra retroflexa*), Crowsfoot Grass, Liverseed Grass, Lovegrass, Pigeon Grass, Summer Grass, Trad (Wandering Jew)	Qld, NSW, WA only	1 L/ha plus 1.46 to 2.2 L/ha Weedmaster* DST*	Apply as an over-the-top or directed spray application when weeds are present. Apply only one application of DUAL GOLD® per crop.

Crops	Weeds Controlled	State	Rate	Critical Comments
Fallows prior to planting crops and as described under Crop Replanting Interval	Awnless Barnyard Grass, Barnyard Grass, Common Sowthistle. Crowsfoot Grass, Feathertop Rhodes Grass, Liverseed Grass, Love Grass, Pigeon Grass, Stinkgrass and Summer Grass	QLD, NSW only	1 to 2 L/ha	Apply in fallows, following crop harvest but not closer to planting the following crop than the period specified under **CROP REPLANTING INTERVA**L. Minimum replanting intervals may apply to crops following DUAL GOLD® application. Best results are achieved when applied to an even, weed-free soil and before weeds begin germination. Sufficient rainfall or irrigation is required after application and prior to weed emergence to move the herbicide into the zone of emerging weeds and allow uptake. DUAL GOLD® will not control emerged weeds and must be used with the recommended rate of a knockdown herbicide where weeds are present. Cultivation following application may reduce pre-emergence weed control.
Fallows	Awnless Barnyard Grass, Barnyard Grass, Common Sowthistle, Crowsfoot Grass, Feathertop Rhodes Grass, Liverseed Grass, Love Grass, Pigeon Grass, Stinkgrass and Summer Grass	QLD, NSW only	**Applied as a split application:** 1.0 to 1.5 L/ha followed by 0.5 to 1.0 L/ha	**DO NOT apply more than 2 L/ha in a single season.** Apply to provide extended residual control in sites with a history of effective weed control and where reduced weed seed loads exist. Avoid splitting the herbicide dose in situations where substantial weed numbers are expected or on heavy soils and apply DUAL GOLD® at the maximum label rate. Apply the initial application pre-emergent to weeds before or at planting or fallow establishment. Apply a follow-up application within 4 weeks of the initial application. DUAL GOLD® will not control emerged weeds and both the initial application and the follow-up application must be applied before weeds emerge. Sufficient rainfall is required following application and prior to weed emergence to move the herbicide into the zone of emerging weeds and allow uptake.
Green Beans and Navy Beans	Blackberry Nightshade, Common Sowthistle, Fat Hen, Feathertop Rhodes Grass, Potato Weed, Barnyard Grass, Crowsfoot Grass, Liverseed Grass, Love Grass, Pigeon Grass, Stink Grass, Summer Grass and **suppression** of Pigweed	All States	1.5 to 2 L/ha	**Pre-emergent:** Apply before, at or immediately after planting and before crops and weeds have germinated. Rain or irrigation is necessary within 10 days of spraying to thoroughly wet the top 3 to 4 cm of soil. If rain does not occur or irrigation is unavailable, incorporate chemical to a depth of 3 to 4 cm. DO NOT throw untreated soil onto treated areas as this will reduce weed control. DO NOT use on soils where the sand content exceeds 70%. Use lower rate where sand content is within the range of 50% to 70%.

Crops	Weeds Controlled	State	Rate	Critical Comments
Green Beans and Navy Beans *continued*	Powell's Amaranth	All States	2 L/ha	**Pre-emergent:** Apply before, at or immediately after planting and before crops and weeds have germinated. For sufficient Amaranth control, the DUAL GOLD® application should be followed by 1 L/ha Basagran* (2 leaf Amaranth) or 2 L/ha Basagran (4 leaf Amaranth). Rain or irrigation is necessary within 10 days of spraying to thoroughly wet the top 3 to 4 cm of soil. If rain does not occur or irrigation is unavailable, incorporate chemical to a depth of 3 to 4 cm. DO NOT throw untreated soil onto treated areas as this will reduce weed control. DO NOT use on soils where the sand content exceeds 70%. Use lower rate where sand content is within the range of 50% to 70%.
Maize, Sweet Corn	Awnless Barnyard Grass, Barnyard Grass, Crowsfoot Grass, Feathertop Rhodes Grass, Liverseed Grass, Love Grass, Pigeon Grass, Stinkgrass, Summer Grass, Trad (Wandering Jew)	All States	1 to 2 L/ha	**Pre-emergent:** Apply before, at or immediately after planting and before crops and weeds have germinated using a low volume boom sprayer applying 60 to 120 L mixture/ha. Rain or irrigation is necessary within 10 days of spraying to thoroughly wet the top 3 to 4 cm of soil. If rain does not occur or irrigation is unavailable, incorporate chemical to a depth of 3 to 4 cm. Use rates towards the higher end of the range on heavy soils or where a heavy grass population is expected. DO NOT throw untreated soil onto treated areas as this will reduce weed control. DO NOT use on soils where the sand content exceeds 70%. Use lower rate where the sand content of the soil is within the range of 50% to 70%.
	Blackberry Nightshade, Caltrop, Common Thornapple, Mintweed, Needle Burr, Noogoora Burr, Pigweed, Redroot Amaranth, Slim Amaranth		1 to 2 L/ha plus 2.5 L/ha of atrazine (500 g/L)	
Mungbeans	Awnless Barnyard Grass, Barnyard Grass, Common Sowthistle. Crowsfoot Grass, Feathertop Rhodes Grass, Liverseed Grass, Love Grass, Pigeon Grass, Stinkgrass and Summer Grass	All States		**All Mungbean use patterns:** Rain or irrigation is necessary within 10 days of spraying to thoroughly wet the top 3 to 4 cm of soil and ensure herbicide activation. DUAL GOLD® will not control emerged weeds and both the initial application and the follow-up application must be applied before weeds emerge. Sufficient rainfall is required following each application and prior to weed emergence to move the herbicide into the zone of emerging weeds and allow herbicide activation and uptake. DO NOT throw untreated soil onto treated areas as this will reduce weed control.
			Pre-emergent: 1 to 2 L/ha	**Pre-emergent:** Apply up to 4 weeks before sowing, immediately after planting and before the crop and weeds have germinated. Use rates toward the higher end of the range on heavy soils or where a heavy grass population is expected.

Crops	Weeds Controlled	State	Rate	Critical Comments
Mungbeans *continued*	Awnless Barnyard Grass, Barnyard Grass, Common Sowthistle. Crowsfoot Grass, Feathertop Rhodes Grass, Liverseed Grass, Love Grass, Pigeon Grass, Stinkgrass and Summer Grass	All States	**Post-emergent:** 1 L/ha	**Post-emergent:** Apply before the weeds have germinated up to 4 leaf crop stage. Post-emergent use of DUAL GOLD® may result in necrotic spotting on leaf tissue exposed to the spray application. New growth with be unaffected and no biomass impact is expected. DO NOT apply after the 4 leaf growth stage
			Applied as a split application: 1.0 to 1.5 L/ha followed by 0.5 to 1.0 L/ha	**Split application: DO NOT apply more than 2 L/ha in a single season.** DO NOT apply after the 4 leaf growth stage Apply to provide extended residual control in sites with a history of effective weed control and where reduced weed seed loads exist. Avoid splitting the herbicide dose in situations where substantial weed numbers are expected or on heavy soils and apply DUAL GOLD® at the maximum label rate. Apply up to 4 weeks before sowing, immediately after planting and before the crop and weeds have germinated or post-emergent up to 4 leaf crop stage. Post-emergent use of DUAL GOLD® at the higher rate may result in necrotic spotting on leaf tissue exposed to the spray application.
Pastures Trikkala Sub-Clover, Haifa White Clover, Maral Persian Clover, Paradana Balansa Clover, Redquin Red Clover, Phalaris and Cocksfoot	Toad Rush	NSW, Vic, SA, Tas, WA, NT, only	200 to 250 mL/ha	Apply at or immediately after planting and before pasture species and weeds emerge. Application should be made to a moist seed bed. Sufficient rain to thoroughly wet the top 3 to 4 cm of soil should occur within 10 days after spraying. Use rates towards the higher end of the range in wet areas or when longer residual control is required. DO NOT use in soils where prolonged waterlogging is anticipated otherwise unacceptable crop retardation will occur. Avoid damage to pasture species by avoiding overlaps and double spraying in headlands where White, Persian and Balansa Clovers are sown and if the 250 mL rate is being used.
Peanuts	Awnless Barnyard Grass, Barnyard Grass, Crowsfoot Grass, Feathertop Rhodes Grass, Liverseed Grass, Love Grass, Pigeon Grass, Stinkgrass, Summer Grass, Trad (Wandering Jew)	Qld, NSW, WA, NT only	1 to 2 L/ha	Apply before pre-irrigation where soil crusting is likely. Apply before, at or immediately after planting and before weeds germinate. Rain or irrigation is necessary within 10 days of spraying to thoroughly wet the top 3 to 4 cm of soil. If rain does not occur or irrigation is unavailable, incorporate chemical to a depth of 3 to 4 cm. DO NOT throw untreated soil onto treated areas as this will reduce weed control. Use rates toward the higher end of the range on heavy soils or where a heavy grass population is expected.

Crops	Weeds Controlled	State	Rate	Critical Comments
Rhubarb	Annual Ryegrass, Apple of Peru, Awnless Barnyard Grass, Blackberry Nightshade, Chickweed (**suppression**), Common Cotula, Common Sowthistle (**suppression**), Crowsfoot Grass, Deadnettle, Fat Hen, Feathertop Rhodes Grass, Fiddle Dock, Fierce Thornapple (**suppression**), Italian Ryegrass, Liverseed Grass, Love Grass, Nettleleaf Goosefoot (**suppression**), Pigeon Grass, Pigweed (**suppression**), Potato Weed, Redroot Amaranth (**suppression**), Shepherd's Purse, Stinging Nettle, Summer Grass, Twiggy Turnip (**suppression**), Winter Grass, Wireweed (**suppression**), Yellow Weed	All States	1 to 2 L/ha	**Pre-emergent:** Apply one (1) application only, immediately before or after transplanting crowns and before weeds have germinated. Sufficient irrigation to wet the soil through the weed zone should be applied within 24 hours. Further weed germination may occur following re-hilling due to exposure of untreated soil.
Spinach, Silverbeet			1.5 to 2 L/ha	Apply with a boom spray, in a minimum of 60 L water/ha. Apply immediately before, at, or immediately after sowing. If transplanting the crop, apply immediately before transplanting. DO NOT apply herbicide over the transplants. Crop retardation may occur where the herbicide is used on soils that have low organic matter content and which contain more than 60% of fine sand and silt. On these soil types, use rates towards the low end of the rate range, and apply to moist soil. Irrigation after application on these types of soils should be limited to no more than 25 mm. Also, on these soils, DO NOT use more than once per year (as on product label for Brassicas). For other relevant instructions, see Broccoli (etc) section of Directions for Use table.
Spring Onions (*Allium fistulosum*) **and Shallots** (*Allium cepa. var. aggragatum*)			1 L/ha	**Direct Seeded Crops:** Apply one application before, at or immediately after planting and before crops and weeds have germinated. **Transplanted Crops:** Apply one application before planting and before weeds have germinated.
Soybeans, Sunflowers	Awnless Barnyard Grass, Barnyard Grass, Crowsfoot Grass, Feathertop Rhodes Grass, Liverseed Grass, Love Grass, Pigeon Grass, Stinkgrass, Summer Grass, Trad (Wandering Jew)	All States	1 to 2 L/ha	Apply before pre-irrigation where soil crusting is likely. Apply before, at or immediately after planting and before weeds germinate. Rain or irrigation is necessary within 10 days of spraying to thoroughly wet the top 3 to 4 cm of soil. If rain does not occur or irrigation is unavailable, incorporate chemical to a depth of 3 to 4 cm. DO NOT throw untreated soil onto treated areas as this will reduce weed control. Use rates toward the higher end of the range on heavy soils or where a heavy grass population is expected.

Crops	Weeds Controlled	State	Rate	Critical Comments
Sorghum treated with CONCEP® II or EPIVIO® C Sorghum Seed Safener	Awnless Barnyard Grass, Barnyard Grass, Crowsfoot Grass, Feathertop Rhodes Grass, Liverseed Grass, Love Grass, Pigeon Grass, Stinkgrass, Summer Grass, Trad (Wandering Jew)	Qld, NSW, WA, NT only	1 to 2 L/ha	**All Sorghum use patterns:** DO NOT apply to Sorghum seed not previously treated with CONCEP® II or EPIVIO® C Sorghum Seed Safener. Rain or irrigation is necessary within 10 days of spraying to thoroughly wet the top 3 to 4 cm of soil and ensure herbicide activation. If applying prior to crop planting and sufficient rainfall does not occur or irrigation is unavailable, incorporate chemical to a depth of 3 to 4 cm. **Pre-emergent:** Apply before, at or immediately after planting and before crops and weeds have germinated using a low volume boom sprayer applying 60 to 120 L mixture/ha. Use rates toward the higher end of the range on heavy soils or where a heavy grass population is expected. DO NOT throw untreated soil onto treated areas as this will reduce weed control. Use the lower rate on sandy soils.
	Blackberry Nightshade, Caltrop, Common Thornapple, Mintweed, Needle Burr, Noogoora Burr, Pigweed, Redroot Amaranth, Slim Amaranth		1 to 2 L/ha plus 2.5 L/ha of atrazine (500 g/L)	
	Awnless Barnyard Grass, Barnyard Grass, Crowsfoot Grass, Feathertop Rhodes Grass, Liverseed Grass, Love Grass, Pigeon Grass, Stinkgrass, Summer Grass		**Applied as a split application:** 1.0 to 1.5 L/ha up to four weeks prior to sowing, followed by 0.5 to 1.0 L/ha at or immediately after planting <u>or</u> 1.0 to 1.5 L/ha at or immediately after planting, followed by 0.5 to 1.0 L/ha post crop emergence	**Split application:** **DO NOT exceed a total of 2.0 L/ha in a single growing season.** DO NOT apply after the 6 leaf growth stage. Apply to provide extended residual control in sites with a history of effective weed control and where reduced weed seed loads exist. Avoid splitting the herbicide dose in situations where substantial weed numbers are expected or on heavy soils and apply DUAL GOLD® at the maximum label rate. Apply a follow-up application within 4 to 6 weeks of the initial application, with the final application occurring no later than the 6-leaf stage of the crop. DUAL GOLD® will not control emerged weeds and both the initial application and the follow-up application must be applied before weeds emerge. Sufficient rainfall is required following each application and prior to weed emergence to move the herbicide into the zone of emerging weeds and allow herbicide activation and uptake.

Crops	Weeds Controlled	State	Rate	Critical Comments
Sugar Cane plant and ratoon	Awnless Barnyard Grass, Barnyard Grass, Bellvine, Billygoat Weed (Blue Top), Blackberry Nightshade, Caltrop, Common Sowthistle (Milk Thistle), Common Thornapple, Crowsfoot Grass, Feathertop Rhodes Grass, Green Summer Grass, Guinea Grass, Liverseed Grass, Mexican Clover, Mintweed, Needle Burr, Passionfruit vines, Pigeon Grass, Pigweed, Redroot Amaranth, Square Weed, Star of Bethlehem, Summer Grass, Wild Rose	Southern Qld (Bundaberg South) and Northern NSW only	1.1 to 1.45 L plus 2.7 to 3.6 L atrazine (500 g/L) or 1.5 to 2 kg of GESAPRIM® Granules (900 g/kg)	Apply as a pre- or post-emergent application to crop. Apply once per year only. Application should be made to a moist soil before weeds and grasses have germinated. If conditions remain dry for a period of 10 days after spraying, irrigation or a shallow cultivation (2.5 cm) may assist results. DO NOT throw untreated soil onto treated areas as this will reduce weed control. In Southern Queensland and Northern New South Wales, use rates toward the higher end of the range on heavy soils where a high grass population is expected. In Northern Queensland, application must be made to moist soil and rainfall or irrigation should occur within 24 hours of application. Use rates toward the higher end of the range where high Green Summer Grass population is expected. Where broadleaf weeds and grasses have emerged and are in the 2 to 4 true leaf stage, knockdown herbicides such as ametryn (500 g/L), a flowable diuron (500 g/L) at 1 L/ha or GRAMOXONE® 360 Pro (0.7 L/ha), applied as a post directed spray, can be added. Where broadleaf weeds only have emerged and are at the 2 to 4 true leaf stage, then a 500 g/L 2,4-D amine at 2 L/ha should be added. In all cases, add a non-ionic surfactant. Follow all instructions and restrictions on the GESAPRIM® labels. **DIURON TANK MIXES:** Read and follow all label directions including restraints, spray drift restraints, mandatory no-spray zones, Critical Comments, withholding periods, regional use restrictions and safety directions for the tank mix products.
		Northern Qld only (Mackay North)	1.45 to 1.8 L plus 3.6 to 4.5 L atrazine (500 g/L) or 2 to 2.5 kg of GESAPRIM® Granules (900 g/kg)	
Sweet Potatoes	Barnyard Grass, Black Pigweed, Crowsfoot Grass, Feathertop Rhodes Grass, Liverseed Grass, Love Grass, Pigeon Grass, Potato Weed, Red Shank, Stinkgrass, Summer Grass, **Suppression** of Evening Primrose	Qld, NSW, Vic, SA, NT only	1.5 L/ha	**Pre-emergent:** Apply immediately after transplanting before weeds have germinated. Sufficient irrigation to wet the soil through the weed zone should be applied within 24 hours. Further weed germination may occur following rehilling due to exposure of untreated soil.

Crops	Weeds Controlled	State	Rate	Critical Comments
Tobacco transplanted in field	Barnyard Grass, Feathertop Rhodes Grass, Love Grass, Pigeon Grass, Stinkgrass, Summer Grass and **suppression** of Common Thornapple (*Datura stramonium*) and Nightshades (*Solanum sarrachoides* and *S. gigrum*)	NSW, Vic only	1.5 L/ha	Apply by low volume boom spray prior to transplanting and before weeds and grasses germinate. The seed bed should be fine and free of large clods. Thoroughly incorporate within 7 days of spraying using a rotary hoe or similar equipment set to work to a depth of 7.5 cm. DO NOT use disc cultivating implements. An even coverage of the herbicide is essential. The herbicide needs to be activated by moisture through to the top 3 to 4 cm of soil at the time of application or within 4 days of incorporation. This can be achieved by applying to moist soil, irrigating soon after incorporation or irrigating immediately after transplanting. Avoid boom overlaps as double treatment may cause growth suppression. Temporary crop retardation may be observed when soil conditions remain dry after application or where Black Root Rot may occur.
Wheat, Barley, Oats, Triticale	Toad Rush	NSW, Vic, Tas, SA, WA, ACT only	150 to 250 mL/ha	Apply at or immediately after planting and before crop and weeds emerge. Application should be made to a moist seed bed and sufficient rain to thoroughly wet the top 3 to 4 cm of soil should occur within 10 days after spraying. Use rates toward the higher end of the range in wet areas or when longer residual control is required or in situations where crop competition is minimal. When mixed with LOGRAN® 750, apply pre-sowing and incorporate by sowing operation (for wheat only). See GENERAL INSTRUCTIONS for use in Zero-Tillage Systems.

NOT TO BE USED FOR ANY PURPOSE, OR IN ANY MANNER, CONTRARY TO THIS LABEL UNLESS AUTHORISED UNDER APPROPRIATE LEGISLATION